

Physics Week 9 Development Run Note

Baseline vs candidate v2, text-only, DeepSeek dev split

Reading guide. This note records development evidence only. Candidate v2 improved over the baseline on the 8-student development split and is now frozen for held-out evaluation. The held-out test split has not been run, so this is not a final accuracy claim.

Experiment State

Validation

8/8 + 8/8

baseline and candidate passed

Exact agreement

85.4%

candidate v2 dev split

Total MAE

0.75

candidate v2, points

Held-out test

Not run

candidate frozen

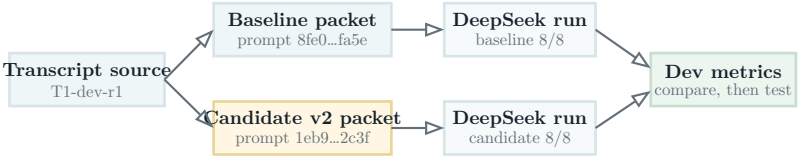
Protocol decision. Candidate v2 is frozen in CANDIDATE-V2-FREEZE.md. The next step is held-out packet preparation without changing prompts.

Data boundary. Student data and model outputs remain under ignored `Data/`. This Git record stores paths, hashes, commands, and aggregate metrics only.

Reproducibility Anchors

Anchor	Value
Record directory	experiments/records/physics-week9-baseline-candidate-v2-run
Branch	codex/physics-week9-baseline-candidate-v2-run
Model-run commit	95622f0aead87187c6410ec0a38ba94cb2866dee
Report input record	DEV-METRICS-STRICT-SCHEMA.md
Course / assessment	physics / week9
Split	development, 8 anonymous students
Provider / model	deepseek / deepseek-v4-pro
Endpoint	https://api.deepseek.com
Input mode	text-only, pilot-derived automatic transcript
Temperature / response format	0 / json_object
Typst	0.15.0

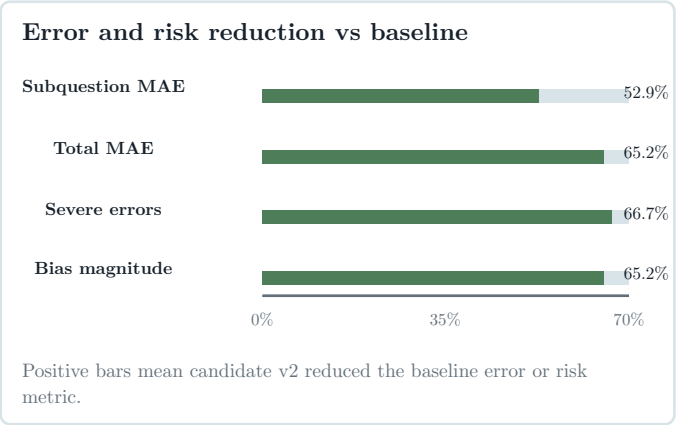
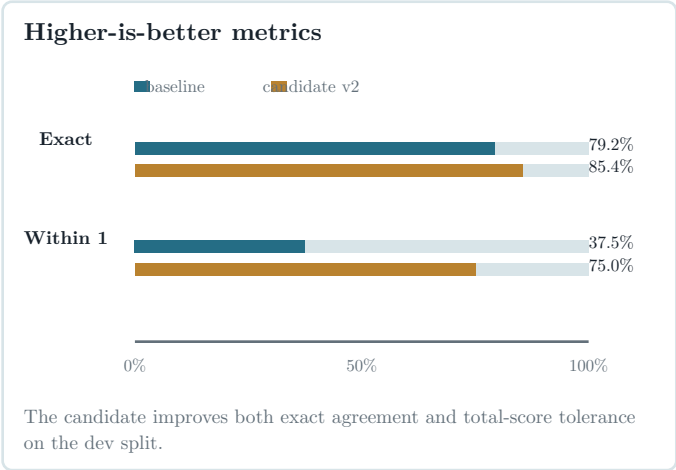
Prompt Packet Lineage



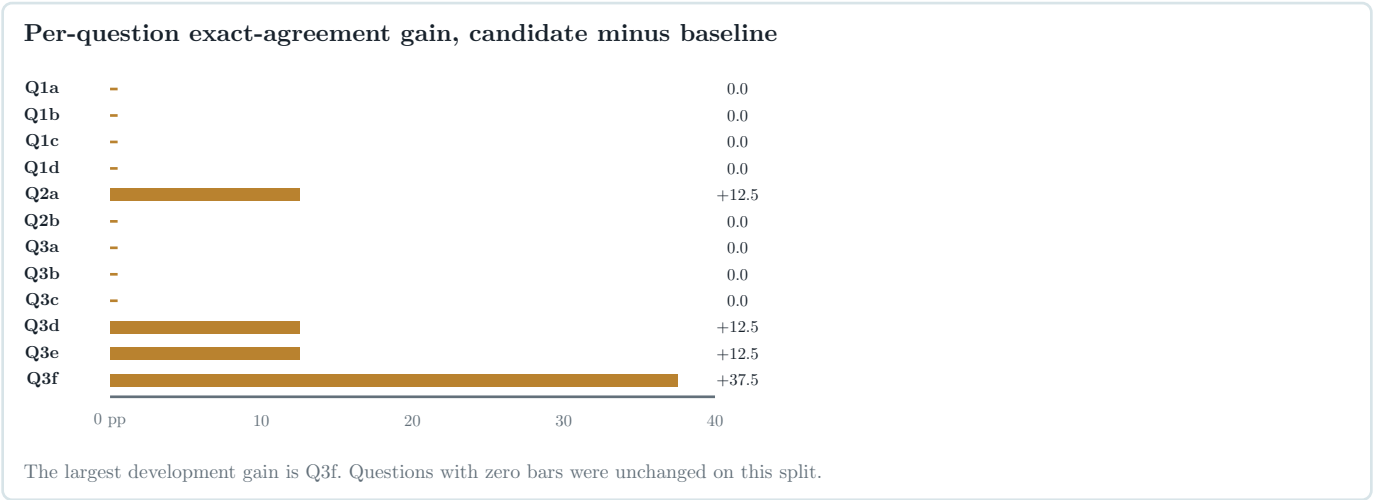
Both packets use the same transcript hash 30e45836b26f6c05d0a55c2e436d5ace7078d01ae86932c3c64b27ad14e24cf8 and rubric hash a02e0531b3d78590c66d32e76eba170d2e0400e3e4a1c60436f4f5c2c8e93b21.

Development Results

Metric	Baseline	Candidate v2	Direction
exact agreement	0.7917	0.8542	candidate better
macro accuracy	0.7917	0.8542	candidate better
subquestion MAE	0.2214	0.1042	candidate better
total score MAE	2.1563	0.7500	candidate better
within 1 point rate	0.3750	0.7500	candidate better
severe error rate	0.3750	0.1250	candidate better
mean signed error	-0.1797	-0.0625	candidate less biased



Where Candidate v2 Changed The Most



Prompt And Packet Registry

This PDF uses short hashes for layout. Full SHA-256 values are recorded in DEV-METRICS-STRICT-SCHEMA.md.

Role	Prompt source	Prompt hash
baseline	experiments/records/.../prompts/grade_standard_v1_strict_schema.txt	8fe0...fa5e
candidate v2	experiments/records/.../prompts/grade_candidate_v2_strict_schema.txt	1eb9...2c3f

Role	Packet path	Packet hash
baseline	Data/physics/benchmark/text_packets/physics-week9-baseline-text-strict-schema/G1-dev-r1	e556...2601
candidate v2	Data/physics/benchmark/text_packets/physics-week9-candidate-v2-text-strict-schema/G1-dev-r1	8987...ed6d

Run Outputs Used For Metrics

Role	Run directory	Validation
baseline	Data/physics/benchmark/runs/physics-week9-baseline-candidate-v2/deepseek-baseline-text-G1-dev-r1-strict-schema	passed, 8/8
candidate v2	Data/physics/benchmark/runs/physics-week9-baseline-candidate-v2/deepseek-candidate-text-G1-dev-r1-strict-schema	passed, 8/8

Local artifact	Hash
dev-metrics-strict-schema.json	68d4...e613
dev-student-total-errors-strict-schema.csv	3e11...b176

Reproduction Commands

Run from the repository root. The API key must be supplied through DEEPSEEK_API_KEY; never write the key value into Git, reports, or shell history.

Step	Command record
1	python -m benchmark.core.cli check-run-readiness ... using baseline-plan.json and candidate-v2-plan.json
2	python -m benchmark.core.cli run-model-packet ... with the baseline strict-schema packet
3	python -m benchmark.core.cli run-model-packet ... with the candidate v2 strict-schema packet

The exact multiline commands are stored in STRICT-SCHEMA-RERUN.md, and the user-facing DeepSeek key setup instructions are stored in EXTERNAL-API-DATA-GATE.md.

Freeze Gate Before Held-Out Test

Gate	Status	Evidence
Development validation	passed	both strict-schema runs validated 8/8
Prompt packet record	passed	prompt, packet, text-source, and rubric hashes are recorded
Candidate v2 freeze	passed	CANDIDATE-V2-FREEZE.md records the frozen prompt
English rubric rule	passed	model-facing prompts and rubrics are English
External API data gate	passed	anonymous data approved for DeepSeek public API by supervisor
Held-out test	not run	build and dry-run held-out packets first

Limitations

- Development split only; 8 anonymous students.
- Text-only input; this run does not test multimodal image grading.
- Text source is a pilot-derived automatic transcript.
- Gold reference status is single primary rater, not adjudicated.
- The failed baseline attempts are retained under ignored Data/ for audit, but are not used for metrics.
- This note supports held-out preparation for frozen candidate v2; it is not a final cross-course conclusion.